X IBM Quantum

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Quantum-Assisted Traffic Signal Optimization Using QAOA

Hackathon Track | Team NULL POINTER | Harshit Kumar (Lead), Himansu Rauniyar, Aditya Pattar | CSE, SIT Tumkur

Introduction & Motivation

Background: Congestion wastes time, fuel and clean air, yet most city signals still run fixed timers that ignore real queues.

Problem Statement: Timing N signals over T slots is combinatorial: there are $2^{N \cdot T}$ possible plans, re-decided every 30 s.

Objective: Solve signal timing with QAOA in Qiskit and test it as a live controller against classical methods.

Research Gap / Existing Challenges

- Today's signals optimise each junction alone.
- Quantum traffic work targets routing [3, 4].
- QAOA is rarely tested as a live controller.

System Architecture / Implementation

1. **Circuit:** H → RZ/RZZ cost → RX mixer, × p layers.
2. **Scale:** 4–16 qubits, p = 1–4; 8 qubits at p = 2: 80 CNOTs, depth 75.
3. **Stack:** Qiskit 2.x, Aer noise model, COBYLA + INTERP [2].
4. **Readout:** 1024 shots → cheapest plan applied.

Theoretical Model / Algorithm

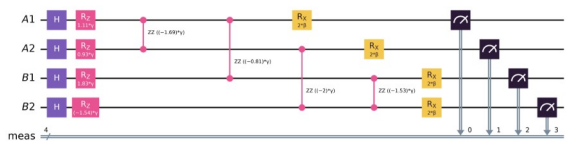
QUBO cost: $E(x) = \Sigma_{i,t} (Q^2_{NS} + Q^2_{EW}) + w \cdot switches$

Ising Hamiltonian (x = (1−Z)/2):

$H = \Sigma\, h_k Z_k + \Sigma\, J_{kl} Z_k Z_l$

QAOA state (p layers): $|\gamma,\beta\rangle = \Pi\, e^{-i\beta\Sigma X} e^{-i\gamma H} |+\rangle^{\otimes n}$

Squared queues couple neighbours, so green waves emerge.



Experimental Results

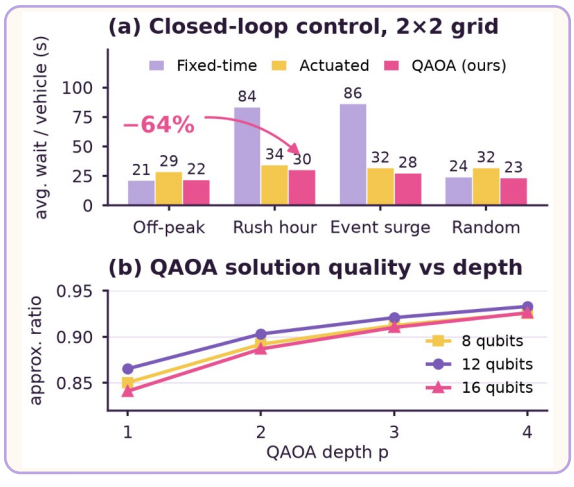


Fig. 1: (a) Avg. wait, 3 seeds; (b) QAOA quality vs depth p.

Proposed Methodology / Framework

Receding-horizon control loop, every 30 s:

1. Encode: one qubit per signal per slot (x = 1 → E–W green).
2. Predict queues for T slots, linear in x; build the QUBO.
3. Map to Ising H_C; train QAOA angles γ, β with COBYLA.
4. Sample 1024 shots; apply the cheapest plan for 30 s.
5. Re-plan with live queues, warm-starting the angles.

Baselines: fixed-time, actuated, simulated annealing, exact.

Applications / Use Cases

- Adaptive signals for smart-city corridors.
- Event and emergency traffic surges.
- A real-world QUBO benchmark for QAOA on IBM Quantum.

Demo video + live app

Conclusion & Key Contributions

1. QAOA found the optimum on every instance ≤ 12 qubits.
2. **−64%** wait vs fixed timers at rush hour (83.5 → 30.2 s).
3. Optimal even at 1% gate noise; quality 0.85 → 0.93.

Honest limit: classical solvers are still faster at this size.

Future Work

1. Run on IBM Quantum hardware via Qiskit Runtime.
2. Split cities into corridors that fit one device.
3. Real data: SUMO and live detector feeds.

Acknowledgement

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References

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2. L. Zhou et al., Phys. Rev. X 10, 021067 (2020).
3. F. Neukart et al., Front. ICT 4, 29 (2017).
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